Voiland Food Pantry

Final Report Draft 3

Mentor: Maynard Poleng Siev, WSU

Team Name: 11-FA25-SP26-F-MARKETING
Members: Jodie Butterworth

CptS 423 Software Design Project II

Spring 2026

TABLE OF CONTENTS - (Estimated)

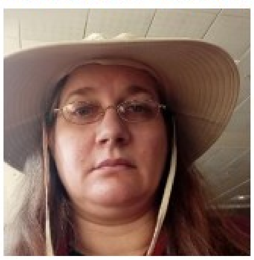
Link to GitHub: https://github.com/WSU-JB5757/WSU_Marketing_423

I. Introduction

As stated in the promotional material*, Washington State University, in an effort to reduce food insecurity, has opened food pantries on campus to provide free food. The development team has created a program to read card data for volunteers and beneficiaries, and provide a database to track the pantry inventory. In addition to the Voiland Food Pantry, the program has potential uses at other campus facilities. The sponsor has requested a rollout plan including all necessary technical documents, testing and specifications of the current program to prepare it for scaling, maintenance, distribution and future revisions by other facilities and development teams. The software program will be tested, documented and prepared for deployment with documentation and revisions from the product development team. The marketing team does not provide code or product development for future program revisions. Instead, it organizes information and plans goals for product rollout and future development.

II. Team Members & Bios

II.1. Jodie Butterworth



Jodie is a senior Computer Science major at Washington State University. Her skills include UI/UX mobile and web development, graphic design, and technical writing. She has prior experience with web development and has worked with WordPress, as well as programming languages such as Python, C/C++/C#, HTML, and CSS. For this project, her responsibilities include front-end development and UI design.

As described in earlier Project Drafts**.

* Introduction from the promotional material abstract for the poster presentation.

** Bio from the team bio section of the development team project report dated 12-7-25.

III.Project Requirements Specification

As agreed in the Client meetings, the scope of the rollout for the marketing project is:***

- Create timeline for product deployment with multiple locations and stakeholders.
- Write a comprehensive user guide for admins, future clients and future teams that includes troubleshooting common problems.
- Create a plan for continued maintenance including hand-off to future project teams.
- Plan marketing meeting and hands-on demonstration with potential future clients.
- Identify potential costs and fees of operating the system.
- Identify quality and performance issues with testing, implement repairs, and record the process and results.
- Create a user's guide for practical daily operation.
- Perform risk analysis on hardware and data secure and notify development team for adjustments.
- Develop a step-by-step plan to save program, data and databases for re-deployment on different hardware. (This will be handled by a tech support expert instead.)

III.1. Project Stakeholders

List of Stakeholders:

- Clients for the Voiland Food Pantry who want a rollout plan for the scanning software to manage inventory and card readers to manage client and volunteer data
- Potential clients in other departments who want to improve their inventory, volunteer and employee data management
- Future users, team project developers and administrators who want clear documentation of the program and how to use and troubleshoot it
- Current development team who want to develop the project with minimal issues.
- Future program users, volunteers and clients who want an easy-to-use program with risk analysis, bug testing, and clear user guides and documentation.

*** As described the meeting minutes for the meeting on 3-5-26, and adjusted in the meeting minutes on 3-26-26

III.2. Use Cases

- U1 As a volunteer of the Voiland Food Pantry, I want to read the user manual so that I know how to run the terminal.
- U2 As an admin of the Voiland Food Pantry, I want to read the comprehensive user guide so that I know how to change the program title.
- U3 As a developer on a future improvement to the program, I want to look at the comprehensive user guide so that I know how to migrate the program and data to different hardware.
- U4 As a new client I want to read the comprehensive user guide so that I know how to operate the database.
- U5 As a developer of a future improvement to the program, I want to read the testing documentation so that I understand the program operation and the testing used.

III.3. Functional Requirements

Based on the objectives from client meetings:***

- FR1 Development team to produce user guide for daily operations that will be edited by marketing.
- FR2 Development to provide information to marketing for a comprehensive user guide for future admins and project teams.
- FR2 Development team to write a report on any costs or fees in operating the product and any foreseen costs of fees of future operations.
- FR3 Development team to provide information to marketing to produce technical specifications and a migration plan.
- FR4 Development team to provide information to marketing to create a list for troubleshooting common problems.
- FR5 Development team to record procedures and results of testing processes and the repairs implemented.

III.4. Non-Functional Requirements

Based on the discussion from client meetings:***

- NR1 Improve communications with development team.
- NR2 Teams to be used in timeline planning.
- NR3 Meetings to be held with potential future clients.

- NR4 Client to be CC in all email communications for the project.
- NR5 Documents to be edited and easy to use.
- NR6 Risk analysis performed on terminal hardware and software in situations of low-supervision and customer self-service.
- Potential user and client feedback to be gathered at demonstration meeting

III.5. Standards

This project is related to a software design project, but is itself not software based. Software testing will be done and compared with other testing results. Teams will be used as project management and communication software. Email is the main communication method.

I have made an app for AI-based project management to create a more complete and interactive rollout planner. It is timeline-based with email reminders and appropriate forms attached to events. It is very basic, flask, sql and python-based program with connection to LLM AI through an API key. It is still experimental, and teams and email will still be used as communication. GitHub will still be used as storage and to turn things in. This will not be a large program, and ChatGPT was used to create documents and troubleshoot.

Aside from that, the rollout documentation mush be edited, complete, easy to use and contain all the required information. Meetings with the client and the instructor may alter the goals as the project moves forward.

Testing Standards were for daily operation and feature functionality. This includes, but was not limited to:

- Verify that if multiple users can be logged on at once, it does not crash the system.

- Verify that navigation is easy to user and intuitive.

- Verify that Barcode code does not break on duplicate entries, repeated use, or QR codes (QR codes do still break it).

- Verify that volunteers can not add impossible hours.

- Verify that volunteers can view the hours they have.

- Allow users with no card or a malfunctioning card to use the Food Pantry.

See separate testing documentation**** and user guide***** for testing information and future possible troubleshooting.

**** As described in the email report for the user testing on 4-10-26.

***** The final user guide includes known issues for future dev teams and standards.

IV. Software Design - From Solution Approach

Cloud-based hosting is analyzed by the development team as a cost-effective deployment alternative for future clients and teams. ChatGPT analysis recommends updates to current system before online deployment. ***** Updates to the current offline system would also increase stability.

Testing completed on 4-9-26 found extensive operating problems. Possibly, this is the result of the recent update. See the testing section for a summary of the testing results.

After risk analysis of the operating environment, additional encryption was added to the card reader data, and the physical hardware will be secured from theft and tampering at the terminal at in the food pantry. A Guest login was added to allow users to access the program if their card can not be used.

For the AI portion, it will be a better timeline organizer with more features for communication and AI assistance incorporated. It will help create emails and forms and email reminders. It relies on LLM online.

IV.1. Architecture Design

The real program is made by the development team for this project.

The smaller AI project will be in python using flask and spl, and will connect email and LLM AI to a timeline with data to process into emails and forms. A zip file of the final draft of this ai program has been uploaded into GitHub, but the real project code for this marketing project is in the related development team project. Additional improvements to the UI and more features could still be added based on the final draft.

IV.2. Data design

The real program is made by the development team for this project.

Since it's a small program, the python program creates a database to save the tasks and create a timeline. It is a small sql of one table located in the program root directory Since it's connected to AI, it will pull needed information from online. For future, it would be more useful if on a phone as well as a laptop. Security would need updating as well.

IV.3. User Interface Design

The real program is made by the development team for this project.

The program is started with terminal and has a simple python-based UI.

IV.4. Constraints

See the project scope under Project Requirements.

V. Test Case Specifications and Results

V.1. Testing Overview****

Testing has been conducted by the development team. The only major errors were in the Firefox UI. User testing continued in the FIZ under the client's supervision. Initial reports were that it was easy to use as a self-service terminal, and has no other major errors.

From a marketing project perspective, the documentation was only a user guide. It only provided enough features to describe basic program operation as a client, but not enough for admins or future teams. The current user guide is more comprehensive, and includes possible problems to test for.

Unfortunately, I have not seen a full testing report by the dev team.

On 4-9-26, I conducted program testing with volunteer users arranged by the client. Several major problems were observed. Results were recorded by the volunteers on feedback forms, and I provided limited direction to ensure thorough testing of all the features. After 4 test users in approximately an hour, the barcode scanner was non-functional as was the admin login. These users did nothing radical or unusual.

The program lacked basic required features and navigation, and it contains unstable bugs that make it unusable after limited normal use. This has since been updated to include back buttons and log out buttons. The program is more stable, but still has stability and security issues. These issues are described in the user guide*****.

V.2. Environment Requirements

The Sloan Building has experienced a string of thefts recently. The terminal should not be left alone and unsecured.

The client has crated a plan to physically secure the terminal.

V.3. Test Results****

As of the email sent to the dev team, client and instructor on 4-9-26, these are the most prominent problems.:

- Ctrl Q needed to exit report prints. (Back button added to fix this.)
- Multiple pages have no way back to home page. (Back button added to fix this.)
- Card scanner doesn't always work. (Guest login added as a word-around.)
- More instructions needed for errors. (Some error messages added.)

**** As described in the email report for the user testing on 4-10-26.

- More instructions needed for user. (Some more text added to the UI.)
- Barcode scanner started showing a JSON error. (Improved by log out button, but still in need of revision.)
- Admin access now broken, could be password. (Admin login working during demo at poster presentation.)
- Volunteers can't view their hours. (Hours are now displayed.)
- Volunteers have no limits on hours number or dates. (Limits have been added.)
- Barcode scanner sometimes doesn't work without JSON error. (Improved, but in need of more revisions.)
- Volunteers sometimes can't update the inventory using the keyboard. (Extension of underlying database/UI problem. Usually works now.)
- No contact information at the bottom of the page. (Still no contact info.)
- Back buttons needed. (Back buttons added.)

The feedback form I gave out was geared toward user acceptance testing, and asks questions geared toward intuitiveness, likelihood of use, and most desired features. Due to the program being so buggy, there were comments about needing Ctrl Q to leave a form, the card scanner not operating randomly and entering items into the inventory with the barcode scanner and by typing on the keyboard becoming impossible.

The previous version of the program seemed more stable even though it lacked features. I have suggested to the dev team that if the current update can not be stabilized, we need to roll back to a previous version and allow the next dev team to take over.

In terms of documentation, the dev team has allowed me GitHub access, but provided no response to the emails that I sent requesting more information so that I can create a more extensive and useful user guide. That said, given the program’s current state, a user guide will not be able to troubleshoot and work-around these problems. I intend to wait for a more stable update to the program to make a user guide.

The update was the same day as the poster presentation. For good or bad, it falls to the future dev team. I have used AI to analyze the code, and combined with known behaviors and issues, I have outlined areas that need further testing and improvement.***** I created a more detailed document specifically for the future that I will send out with the updated report and user guide*****.

The program is improved, and mostly stable for offline terminal use. It could use improvements, as described in the user guide*****. (Too many *?). It must not be deployed online as is.

In terms of documentation, the stakeholder feedback form was created by marketing, and a basic user guide was created by the dev team. A comprehensive user guide with a future work section is made, a known issues document has been made.

VI. Projects and Tools used

I'm using GitHub, Teams, Python, Flask, SQL, word processors and ChatGPT.

VII. Description of Final Prototype

The final project will include a user manual, stakeholder feedback forms, a comprehensive user guide for admins and future teams that includes troubleshooting, an analysis of operating costs for future operations (done by dev team), testing and testing records (dev team, client and marketing), a known issues summary for future work, a timeline with tasks for the rollout and an AI planner to automate some of these tasks.

VIII. Social Responsibility and Broader Impacts

The food pantry provides a valuable community service. Improving their software to more efficiently track volunteer hours, customer use, and product inventory will enable them to do their job more efficiently. A rollout plan makes the transition for the next team who takes over with the project improves easier for them. It also connects with potential clients and provides both current and future clients and administrators the records and instructions they need to better utilize the program. It also analyzes the environment for risks and warns the development team of potential issues.

IX. Product Delivery Status

A booklet for the poster presentation was developed, but it has been merged with the development team presentation.

Initial testing is completed and error-correction for the UI by the development team has begun.

User testing was completed with feedback forms, and several bugs and missing features were identified and reported to the dev team.

The cost-analysis and resulting planning for migration of the database to cloud or other hardware is also being planned by the development team.

The development team sent a simple user guide for testing purposes. The user guide was forwarded to the client for review, and used in testing.

A rollout timeline has also been established on Teams, as per client request. This timeline has been updated as the project changes.

Environment-based risk analysis done and the development team was notified of potential risks. Plans to further encrypt data and physically secure the terminal have been made.

Plans for potential client meeting with stakeholder feedback and project demonstration set for early April. This was handled by the dev team, and I was not notified in time to make adjustments to testing. Again, communication in this project has been horrible and regularly damages forward progress for both marketing and program development.

Final update for the poster presentation, offline terminal stable and improvements added.

ChatGPT analysis of software, and creation of future testing and improvement recommendations in both the user guide and a separate document.

IX.1. Limitations and Recommendations

I do not directly contribute code for the development team. This includes repairing identified problems in user testing. They are responsible for giving me information to make the comprehensive user guide and stakeholder feedback form. I have completed the feedback form, and the user guide with future troubleshooting and improvement recommendations. As a note, I was involved in creating the prototype and in the project planning, but I no longer contribute code or plans.

IX.2. Future Work

I will collect program and testing information and create the comprehensive user guide or a known bugs and issues report for repairs by future dev teams if necessary. (I ended up doing both.) The future deployment of the program to other departments and locations is partially dependent on the poster demonstration recent program repairs. As stated, improvements could be made for using it offline, and improvements must be made for online deployment.

The AI planner adds code to this side project, but is not the proper Voiland Food Pantry project code. I will continue to use email, Teams and GitHub with the planner as an experimental addition, really. A zip of the ai helper is available in GitHub. I plan to modify it again for better features at a future date.

X. Acknowledgements

I thank Maynard Poleng Siev for sponsoring the marketing project. I thank the development team and my instructor for reviewing the marketing project and for putting in the extra work to make it viable.